DIGITAL DOOR ACCESS SYSTEM

USING COMBINATIONAL LOGIC

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INTRODUCTION & OBJECTIVE

Introduction

- Mechanical keys can be lost, copied or picked.
- Many electronic locks are complex and costly for simple needs.
- Our system uses basic gates (AND, OR, NOT) to unlock the door only when the entered 4-bit code matches an authorised code.
- Purely combinational: output depends only on present inputs, with no memory or clock.

Objective

- Design a circuit that unlocks only for authorised codes.
- Derive and minimise the Boolean expression.
- Verify with a truth table and K-map.
- Simulate correct, wrong and edge cases in Logisim.
- Add zone-based control to save energy.

THEORY & LOGIC CIRCUIT

Basic concept

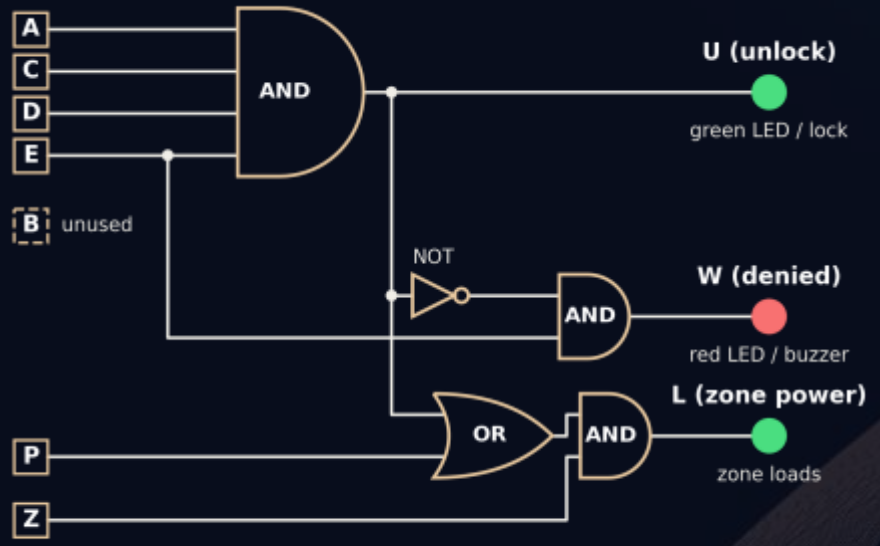
- Combinational logic: output depends only on present inputs.
- Gates used: AND, OR, NOT.
- Authorised codes: 1011 (user) and 1111 (master).

U = E·A·C·D

W = E·U'

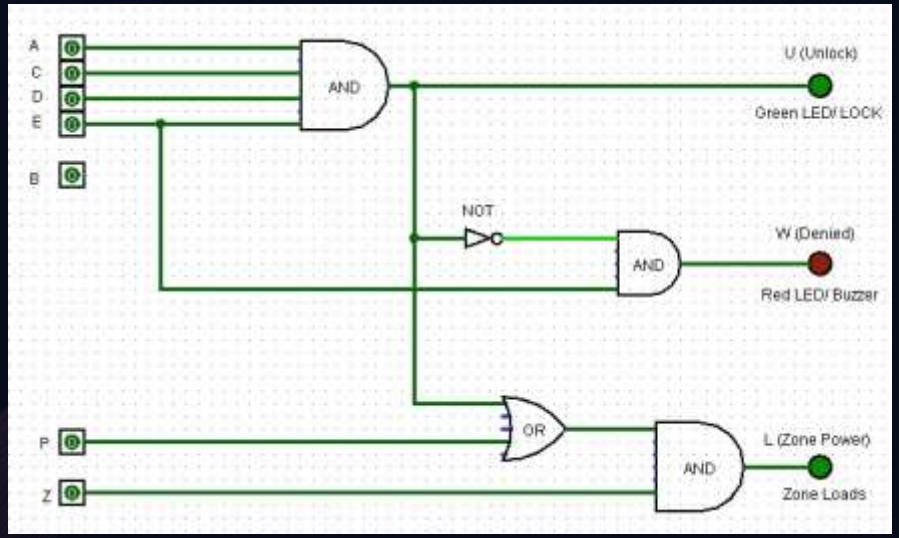
L = Z·(P + U)

K-map: B is a don't-care, so unlock needs just one AND gate.



Wires that cross without a dot are not connected.

CIRCUIT DESIGN IN LOGISIM

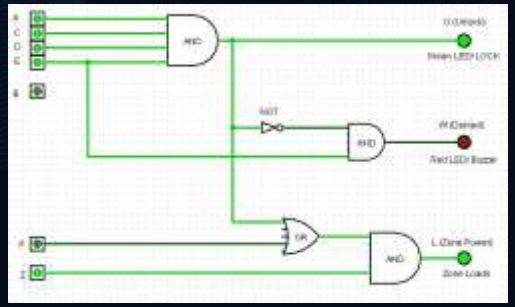


Logisim circuit (starting state: all inputs 0)

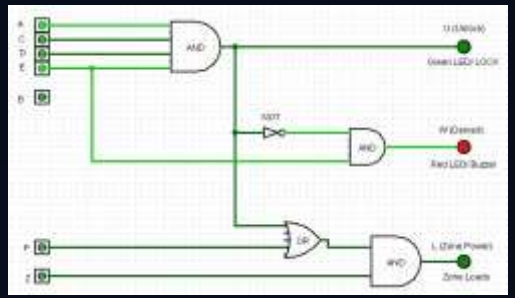
How it works

1. Set the code on A, B, C, D and press Enter (E = 1).
2. AND / NOT gates compare it with 1011 and 1111.
3. Match: U = 1, door unlocks, green LED ON.
4. No match: W = 1, red LED / buzzer ON.
5. Zone: L = Z·(P + U) powers a zone only if enabled and occupied or just unlocked.

SIMULATION RESULTS



Access granted: code 1011, green LED ON



Access denied: code 1000, red LED ON

Input–output combinations

E	Code ABCD	U	W
0	any	0	0
1	1011	**1**	0
1	1111	**1**	0
1	1000	0	**1**
1	1110	0	**1**
1	0000	0	**1**

All 16 codes were tested: only 1011 and 1111 give U = 1.

Zone L = Z·(P + U): Z = 1 with P = 1 gives L = 1. Z = 1, P = 0, U = 0 gives L = 0. Z = 0 always gives L = 0.

CONCLUSION & APPLICATIONS

Conclusion

- Door access system designed and simulated using only combinational logic.
- Truth table and Logisim simulation agree.
- K-map reduced unlock logic to a single AND gate.

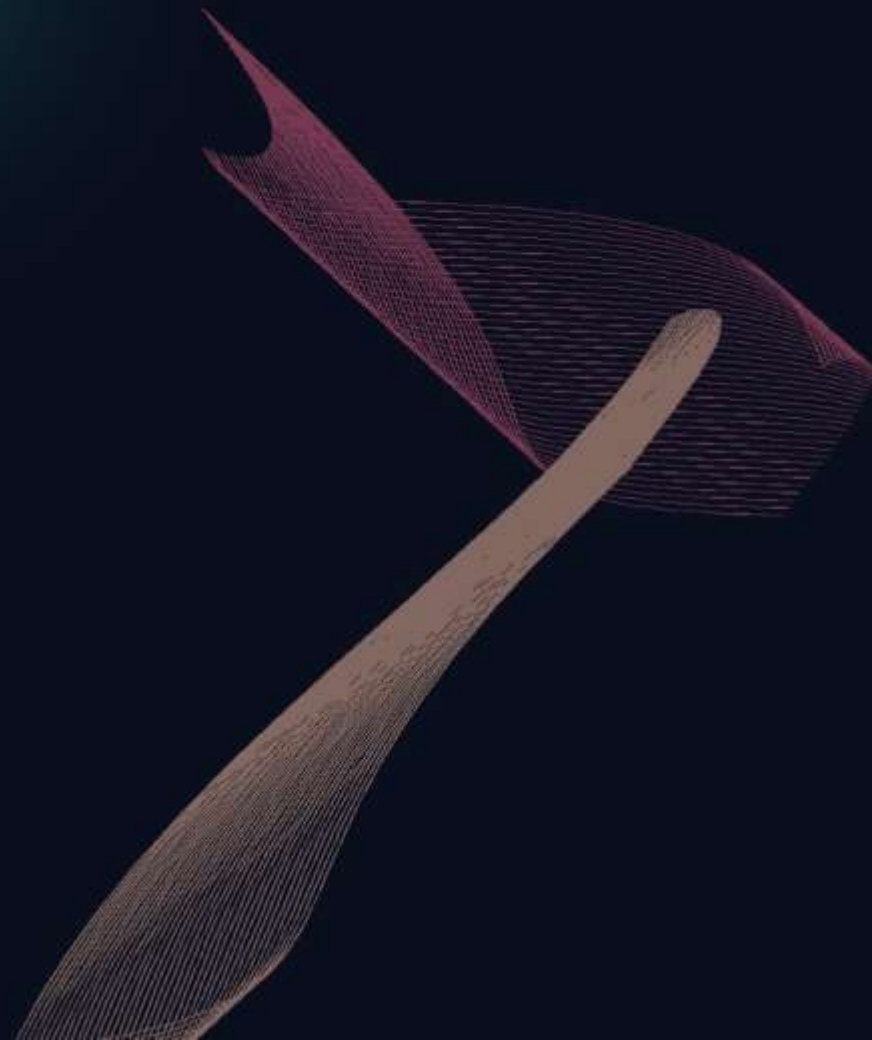
Applications

- Homes, offices and laboratories.
- Server rooms and restricted areas.
- Teaching combinational design.

Improvements

- Keypad encoder for easier entry.
- Flip-flops for wrong-attempt lockout.
- Longer codes; Arduino, RFID or IoT.

THANK YOU



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